representation theory (of Lie groups and Lie algebras)

in physics we usually focus on matrix groups

ex:

- $GL(n,\mathbb{R})$, $GL(n,\mathbb{C})$
- $SL(n,\mathbb{R})$, $SL(n,\mathbb{C})$
- $O(n)$, $O(n,\mathbb{C})$ $\qquad x,y \in \mathbb{C}^n \qquad x\cdot y = x^* y$ conjugate transpose

$\downarrow$

group of transformations that leaves an inner product invariant

for $\mathbb{R}^n$: $x,y \in \mathbb{R}^n$, $x\cdot y = x^t y$

if $A \in GL(n,\mathbb{R})$, A acts on x,y by $x \to Ax$, $y \to Ay$

A changes inner product by $x\cdot y \to (Ax)^t(Ay) = x^t \underbrace{A^t A}_{=\mathbb{1}} y$

$A^t = A^{-1} \in O(n)$

- $U(n) \subseteq GL(n,\mathbb{C})$ $\qquad A \in U(n)$ if $A^* = A^{-1}$

$SO(n)$, $SU(n)$ same as before, with determinant 1

$SO(3,1)$ Lorentz group

$\to$ preserves the Minkowski "inner" product (Minkowski metric)

$$\langle x,y\rangle = -x^0y^0 + x^1y^1 + x^2y^2 + x^3y^3$$

$$SO(3,1) = \{A \in SL(4) \mid \langle Ax, Ay\rangle = \langle x,y\rangle\}$$

$SO(3) \subseteq SO(3,1)$ subgroup of rotations

So what is a representation ?

$\rho : G \to \mathrm{Aut}(V)$ $\qquad$ V complex vector space

$\to$ group of linear operators acting : $V \to V$

group homomorphism

$\to$ we get a linear action of G on V

ρ is also smooth

Lie group: a smooth manifold with a group operation such that group multiplication and inverse operations are smooth

for a matrix Lie group $G \leq GL(n, \mathbb{C})$ which is closed (topologically)

if we fix $g \in G$ the map $m_g : h \in G \to g \cdot h \in G$ smooth

---

To every Lie group we can associate a Lie algebra

For a MLG $G \leq GL(n, \mathbb{C})$ the associated Lie algebra is a real vector space defined by $\mathfrak{g} = \{ X \in \overbrace{M_n(\mathbb{C})}^{n \times n \text{ matrices}} \mid \underbrace{e^{tX}}_{\text{matrix exponential}} \in G \ \forall t \in \mathbb{R} \}$

Lie algebra represents infinitesimal transformations

$$e^{tX} \in G \to e^{tX} = \mathbb{1} + t\underline{X} + o(t^2)$$

$$e(x+y) = e^x e^y$$

Def: a Lie algebra is a vector space $\mathfrak{g}$ with an operation $[\cdot,\cdot] \in \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ (Lie bracket) that must satisfy:

- bilinear
- satisfies Jacobi identity $[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$

- anti-symmetric $[X,Y] = -[Y,X]$

---

simplest example: $M_n(\mathbb{C})$ (as real vector space)

$$[X,Y] = XY - YX$$

---

back to $\mathfrak{g} = \{ X \in M_n(\mathbb{C}) \mid e^{tX} \in G \quad \forall t \in \mathbb{R} \}$

the Lie bracket is $[X,Y] = XY - YX$

- $G = GL(n,\mathbb{C})$ if $X \in M_n(\mathbb{C})$ $e^X \in GL(n,\mathbb{C})$

  $tX \in M_n(\mathbb{C})$ $e^{tX} \in GL(n,\mathbb{C})$

  $\mathfrak{gl}(n,\mathbb{C}) = M_n(\mathbb{C})$

- $G = GL(n,\mathbb{R})$ if $X \in M_n(\mathbb{R})$ $e^X \in GL(n,\mathbb{R})$

  $\mathfrak{gl}(n,\mathbb{R}) = M_n(\mathbb{R})$

- $G = SL(n,\mathbb{C})$

$$\det(e^{tX}) = 1 \quad \forall t \in \mathbb{R}$$
$$\parallel$$
$$e^{t\,tr(X)} = 1 \quad \forall t$$

$\frac{d}{dt}\Big|_{t=0}$ on both sides $\rightarrow$ $tr(X) = 0$

$$\mathfrak{sl}(n,\mathbb{C}) = \{ X \in M_n(\mathbb{C}) \mid tr(X) = 0 \}$$

$$(X^n)^* = (X^*)^n$$

- $G = U(n)$

$$(e^{tX})^* = (e^{tX})^{-1} \quad \forall t \in \mathbb{R}$$
$$e^{tX^*} = e^{-tX} \quad \forall t \in \mathbb{R}$$
$$\Updownarrow$$
$$X^* = -X$$

$$\mathfrak{u}(n) = \{ X \in M_n(\mathbb{C}) \mid X^* = -X \}$$

$$\mathfrak{su}(n) = \{ X \in M_n(\mathbb{C}) \mid X^* = -X, \ tr(X) = 0 \}$$

$\mathfrak{u}(n)$ real vector space because if $X \in \mathfrak{u}(n)$, $\alpha \in \mathbb{R}$

$$(\alpha X)^* = \alpha X^* = -\alpha X = -(\alpha X) \quad \rightarrow \quad \alpha X \in \mathfrak{u}(n)$$

$$(iX)^* = -iX^* = +iX \neq -(iX)$$

$$\mathfrak{o}(n) = \{ X \in M_n(\mathbb{R}) \mid X^t = -X \}$$

$$\mathfrak{so}(n) = \mathfrak{o}(n)$$

$SO(2) \rightarrow$ rotation matrices in the plane

$SO(2) = \{ X \in M_2(\mathbb{R}) \mid X^t = X^{-1}, \det X = 1 \}$

$$\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^t, \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = 1 \right\}$$

$$= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \ ad - bc = 1 \right\}$$

$$= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| a = d, \ b = -c, \ a^2 + b^2 = 1 \right\}$$

$$= \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \middle| a^2 + b^2 = 1 \right\} = \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \middle| \theta \in [0, 2\pi) \right\}$$

$$\mathfrak{so}(2) = \{ X \in M_2(\mathbb{R}) \mid X^t = -X \}$$

$$= \left\{ \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix} \ a \in \mathbb{R} \right\}$$

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta \, x + \sin\theta \, y \\ -\sin\theta \, x + \cos\theta \, y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$(x')^2 + (y')^2 = x^2 + y^2$$

$x_1, x_2 \in \mathbb{R}^2 \qquad x_1 \cdot x_2 = \cos\theta_2 \, \|x_1\| \, \|x_2\|$

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \mathbb{1} + \begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix} + o(\theta^2)$$

$$\cos\theta \approx 1 - \frac{\theta^2}{2}$$

$$\sin\theta \approx \theta - \frac{\theta^3}{3!}$$

$$\exp\left(\theta \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

if $\rho : G \to \mathrm{Aut}(V)$ representation

induces a rep. of its Lie algebra

$\pi : \mathfrak{g} \to \underline{\mathrm{End}(V)}$ → Lie algebra with commutator

→ linear

→ Lie algebra homomorphism : $\pi([X,Y]) = [\pi(X), \pi(Y)]$

if $X \in \mathfrak{g}$ $\quad \pi(X) = \frac{d}{dt} \rho(e^{tX}) \Big|_{t=0}$ $\qquad \left(X = \frac{d}{dt} e^{tX} \Big|_{t=0} \right)$

---

if G is simply connected, one-one correspondence between rep of group and rep of Lie algebra → $\rho(e^{tX}) = e^{t\pi(X)}$

To classify rep of G, we can look at rep of $\mathfrak{g}$ instead

---

<u>Quantum mechanics</u> states of system live in an Hilbert space V

<u>Wigner' theorem</u> → a symmetry in quantum mechanics must act either as a unitary or anti-unitary operator.

→ the action of a symmetry group on V is a rep. of the group on V

---

if $|\psi\rangle \in V$ state of the system

$$\frac{|\langle \psi' | \psi \rangle|^2}{\langle \psi' | \psi' \rangle \langle \psi | \psi \rangle} = \text{probability of measuring state } \psi \text{ in state } \psi'$$

$\rho : G \to \mathrm{Aut}(V)$ is unitary if $\rho(g)^\dagger = \rho(g)^{-1}$

Spinors?

vectors can be characterized by the way they transform under rotations

vectors are objects that live in a certain representation of the rotation group

---

$SO(2)$ acts on $\mathbb{C}^2$ in this way:

$$\begin{pmatrix} u \\ v \end{pmatrix} \in \mathbb{C}^2 \,, \quad \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \in SO(2)$$

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \triangleright \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$ action of $SO(2)$ on $\mathbb{C}^2$

→ this is a rep. (fundamental representation)

→ 2d vectors are elements of the fundamental rep of $SO(2)$

→ scalars are elements of trivial rep of $SO(2)$

→ $\rho : X \in SO(2) \to id \in Aut(\mathbb{C})$

$$\rho(X)\, u = u$$

To find spinors we need to classify reps of $SO(n)$